The spectral theorem, intuitively

Sanjay Suresh

February 2026

The spectral theorem says every real symmetric (or complex Hermitian) matrix can be written as

$$A = Q\,\Lambda\,Q^{\mathsf{T}},$$

an orthonormal basis Q of eigenvectors and a diagonal matrix Λ of real eigenvalues. The geometric reading is the useful one: such a matrix acts as pure stretching along a set of perpendicular axes, with no rotation or shear.

That is why these matrices are everywhere I work. A Hamiltonian is Hermitian, so its eigenvalues are real energies and its eigenvectors are stationary states. A covariance matrix is symmetric, so PCA is just reading off its eigen-axes. Whenever a problem reduces to a symmetric operator, the spectral theorem promises a clean coordinate system in which it becomes diagonal.